A Machine-Checked Obstruction in the Calibration Step of the Switching-by-Weakness Route to P $\neq$ NP

The MeTTapedia formalization effort

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Abstract

Ben Goertzel's manuscript P $\neq$ NP*: A Non-Relativizing Proof via Quantale Weakness and Geometric Complexity* (v12) proposes to separate P from NP by an information-theoretic *quantale* argument. On an efficiently samplable ensemble $\mathcal{D}_m$ of masked random Unique-SAT instances with a Valiant–Vazirani isolation layer, a *Switching-by-Weakness* normal form (Theorem 4.2) reduces every short global decoder, after a short wrapper, to an $O(\log m)$-local per-block rule on the post-switch input $\mathbf{u}_i = (\mathbf{z}, a_i, b)$; sign-invariant neutrality (Theorem 5.1) and template sparsification (Theorem 5.10) then pin each local rule near 1/2, independence across $\Theta(m)$ blocks yields per-program success $2^{-\Omega(t)}$, hence a distributional lower bound $K^{\text{poly}} \geq \eta\, t$, which clashes with the constant-length self-reduction upper bound under P = NP (Proposition 7.2). The load-bearing step is the *calibration*/success-domination link (Theorem 4.2, Corollary 4.6, Lemma 4.8; Appendix A.5, Lemma A.17), which derives the exchangeability of (X_i, Y_i) at a fixed local input $\mathbf{u}$ from the promise-preserving involution T_i. We record a machine-checked finding of the MeTTapedia formalization effort: the involution T_i acts on the post-switch input $\mathbf{u}_i = (\mathbf{z}, a_i, b)$ by $b \mapsto b \oplus a_i$, so it preserves only the *reduced projection* $(\mathbf{z}, a_i)$, and preserves the *full* input $(\mathbf{z}, a_i, b)$ if and only if $a_i = 0$. Consequently the exchangeability at a fixed $\mathbf{u}$ that the calibration step requires holds only on the $a_i = 0$ fiber — a vanishing $2^{-k} = m^{-\Theta(1)}$ fraction of Valiant–Vazirani columns — and the "local comparator dominates the global decoder" calibration is not valid as stated. The structural facts, together with a finite two-point success-domination counterexample, are formalized with no `sorry` in the Lean module `Mettapedia.Computability.PNP.PostSwitchInputObstruction`. We state the scope exactly: what is machine-checked is the *structural* input-preservation fact; the full claim that *no* local comparator dominates a global decoder on this ensemble engages the genuine difficulty and is *not* established by this work. We make no claim about P versus NP, which remains open; the formalization asserts no separation.

1 The problem and Ben's quantale / Switching-by-Weakness route

1.1 The quantale-weakness route

The question is whether P = NP. Ben Goertzel's manuscript P $\neq$ NP*: A Non-Relativizing Proof via Quantale Weakness and Geometric Complexity*, version v12 (hereafter "v12"), proposes a distributional separation built from three interacting ingredients: a compositional *weakness* calculus measuring the shortness of algorithms, the symmetry and sparsity of masked random 3-CNFs, and a genuine *algorithmic switching* statement turning any short polynomial-time decoder into a local per-bit rule on a constant fraction of blocks.

The weakness quantale. v12 measures shortness by the polytime-capped conditional description length

$$\text{weakness}(z \mid y) \;:=\; K^{\text{poly}}(z \mid y),$$

used as the cost object of a quantale: costs add under composition and under independent block product (v12, Lemma A.8). The strategy is to oppose a *global budget* accrued by any short decoder across $t = \Theta(m)$ independent blocks against the constant-length per-block encoder available under a P = NP self-reduction.

The masked-and-isolated ensemble. For a fixed clause density $\alpha > 0$ and $k = c_1 \log m$, a block is sampled (v12, §3.1) as $\Phi = (F^h, A, b)$, where: F is a constant-density random 3-CNF; a fresh mask $h = (\pi, \sigma) \in H_m = S_m \ltimes (\mathbb{Z}_2)^m$ permutes variable names and flips literal signs to give F^h; and a Valiant–Vazirani isolation layer adds a matrix $A \in \{0,1\}^{k \times m}$ with pairwise-independent columns and a δ-biased right-hand side $b \in \{0,1\}^k$. The block distribution $\mathcal{D}_m$ is the law of Φ conditioned on the event $\mathsf{Unq}(\Phi)$ that Φ has a *unique* satisfying assignment $X = x(\Phi) \in \{0,1\}^m$. Writing $a_i := Ae_i$ for the i-th column of A, the pair (a_i, b) is the *VV label* for bit i.

Local inputs and the post-switch interface. A sign-invariant SILS feature vector $\mathbf{z} := \text{feat}(F^h) \in \{0,1\}^{r(m)}$ with $r(m) = O(\log m)$ is published, and v12 defines (Definition 3.10) the *per-bit local input*

$$\mathbf{u}_i(\Phi) \;:=\; \big(\mathbf{z}(F^h),\; a_i = Ae_i,\; b\big) \;\in\; \{0,1\}^{O(\log m)}, \tag{1}$$

together with the local σ-field $\mathcal{L}_i$ it generates.

Switching-by-Weakness. The structural core (v12, Theorem 4.2) is a normal form: for every polynomial-time decoder P of description length $\le \delta t$ there is a short wrapper W with $|W| \le |P| + O(\log m + \log t)$ and a subset $S \subseteq [t]$ with $|S| \ge \gamma t$ such that

$$(P \circ W)(\Phi)_{j,i} \;=\; h_{j,i}\big(\mathbf{u}_i(\Phi_j)\big), \qquad j \in S,\; i \in [m], \tag{2}$$

for Boolean maps $h_{j,i} : \{0,1\}^{O(\log m)} \to \{0,1\}$, each computable in time $\text{poly}(\log m)$. The wrapper is constructed by ERM distillation (Proposition A.5), trained on *symmetrized, back-mapped* surrogate labels rather than on the truth X; the link from the surrogate labels back to truth is the *calibration* Lemma 4.8.

Neutrality, sparsification, and the clash. Two orthogonal mechanisms then force near-randomness of every $\mathbf{u}$-measurable per-bit rule on $\Theta(t)$ blocks: a sign-invariant *neutrality* lemma (Theorem 5.1), $\Pr[X_i = 1 \mid \mathcal{I}] = \frac{1}{2}$ for any sign-invariant view $\mathcal{I}$; and a *template sparsification* theorem (Theorem 5.10) showing that, at radius $r = c_3 \log m$, any fixed local per-bit rule is ε-high-bias on at most $o(t)$ blocks. Combined with single-block lower bounds for tiny ACC^0/streaming decoders and independence across blocks (Theorem 6.7), one obtains a per-program small-success bound $2^{-\Omega(t)}$, hence the tuple incompressibility $K^{\text{poly}}\big((X_1, \ldots, X_t) \mid (\Phi_1, \ldots, \Phi_t)\big) \ge \eta\, t$ with high probability (Theorem 6.8). Under P = NP, a uniform constant-length program recovers each witness by bit-fixing with a USAT decider, so the same quantity is $\le C$ (Proposition 7.2); for large t the two are incompatible, which v12 reads as P $\neq$ NP (Theorem 7.4).

1.2 The calibration crux

Every quantitative step downstream of (2) bounds the success of the *comparator* $P \circ W$ and transfers that bound back to P by *success domination* (v12, Corollary 4.6):

$$\Pr[P(\Phi) = X] \;\le\; \Pr[(P \circ W)(\Phi) = X] \;+\; m^{-\Omega(1)}. \tag{3}$$

The comparator is trained on symmetrized surrogate labels Y_i, so its analysis is only meaningful once one knows that the Bayes-optimal predictor for the surrogate Y_i at a fixed local input $\mathbf{u}$ is *also* optimal for the truth X_i. This is the calibration Lemma 4.8, and its proof rests on a single exchangeability claim. In v12's words (the "Calibration in one line" of §4.1, proved in Appendix A.5):

> *Fix* $\mathbf{u} = (\mathbf{z}, a_i, b)$. *The promise-preserving involution* T_i *bijects* $(X_i = 0, Y_i = y) \leftrightarrow (X_i = 1, Y_i = 1 - y)$ *without changing* $\mathbf{u}$ *or the measure. Thus* $(X_i, Y_i) \mid \mathbf{u}$ *is exchangeable, so* $\Pr[X_i = 1 \mid \mathbf{u}] = \Pr[Y_i = 1 \mid \mathbf{u}] = f_i(\mathbf{u})$, *and the Bayes rule* $h_i^\star(\mathbf{u}) = \mathbf{1}[f_i(\mathbf{u}) \ge \frac{1}{2}]$ *is optimal for both.*

The involution invoked here is the promise-preserving map of v12, Lemma 3.6,

$$T_i:\; (F^h, A, b) \;\longmapsto\; (F^{\tau_i h},\; A,\; b \oplus Ae_i), \tag{4}$$

where $\tau_i \in H_m$ flips only variable i's literal signs. It toggles the witness bit X_i and fixes any sign-invariant view $\mathcal{I}$, which is exactly the content of neutrality. The calibration argument additionally asserts that T_i acts on the *local input* $\mathbf{u}_i = (\mathbf{z}, a_i, b)$ "without changing $\mathbf{u}$."

The detailed proof in Appendix A.5 (Lemma A.17) makes the action on each coordinate of $\mathbf{u}$ explicit. Its Step 1 lists the properties of the same map (4): it "preserves the SILS features $\mathbf{z}$ (which are sign-invariant)," "preserves a_i but *flips* b *by* a_i," and "preserves the uniqueness promise." Step 2 then conditions on a fixed $\mathbf{u} = (\mathbf{z}, a_i, b)$ — a triple whose *third* coordinate is b — and concludes exchangeability of (X_i, Y_i) "at fixed $\mathbf{u}$." These two statements are in tension: a map that flips b by a_i does not fix the triple $(\mathbf{z}, a_i, b)$ unless $a_i = 0$. It is precisely this bridge point that we formalize.

2 The formalized obstruction

We formalize the action of the involution (4) on the post-switch input (1), isolating exactly which part of $\mathbf{u}$ is invariant. The development is elementary and carries no `sorry`; it lives in the Lean module `Mettapedia.Computability.PNP.PostSwitchInputObstruction`, over a small bit-vector layer in which `BitVec` n is a Boolean function on $\mathrm{Fin}\, n$, `zeroVec` is the all-false vector, and the Valiant–Vazirani right-hand-side update is the coordinatewise exclusive-or

```
def vvToggle {n : Nat} (a b : BitVec n) : BitVec n :=
  fun i => Bool.xor (b i) (a i)
```

which is the XOR $b \oplus a$ appearing in (4).

2.1 The post-switch input and the involution's action

The post-switch input $\mathbf{u} = (\mathbf{z}, a_i, b)$ of (1) is the structure `PostSwitchInput`; the involution's induced action `tiInputMap` is exactly (4) read on $\mathbf{u}$ (the SILS coordinate $\mathbf{z}$ held fixed by sign-invariance, the column a held fixed, and the VV right-hand side b toggled); and `invariantProjection` is the coordinate one obtains by *dropping* b.

```
structure PostSwitchInput (Z : Type*) (k : ℕ) where
  z : Z
  a : BitVec k
  b : BitVec k

/-- The exact local-input action induced by the paper's involution `T_i`. -/
def tiInputMap {Z : Type*} {k : ℕ} (u : PostSwitchInput Z k) : PostSwitchInput Z k :=
  ⟨u.z, u.a, vvToggle u.a u.b⟩

/-- The obvious `T_i`-invariant projection obtained by dropping the VV
right-hand side. -/
def invariantProjection {Z : Type*} {k : ℕ} (u : PostSwitchInput Z k) : Z × BitVec k :=
  (u.z, u.a)
```

The reduced projection $(\mathbf{z}, a_i)$ is always T_i-invariant. This is the part of $\mathbf{u}$ that the calibration argument may legitimately condition on.

```
@[simp] theorem invariantProjection_tiInputMap
    {Z : Type*} {k : ℕ} (u : PostSwitchInput Z k) :
    invariantProjection (tiInputMap u) = invariantProjection u := by
  rfl
```

The full input $(\mathbf{z}, a_i, b)$, however, is fixed by T_i exactly on the zero-column fiber. The pivot lemma is that toggling b by a returns b iff a is the zero vector,

```
lemma vvToggle_eq_self_iff_zero {n : ℕ} (a b : BitVec n) :
    vvToggle a b = b ↔ a = zeroVec
```

from which the full-input preservation criterion follows immediately:

```
theorem tiInputMap_eq_self_iff_zeroColumn
    {Z : Type*} {k : ℕ} (u : PostSwitchInput Z k) :
    tiInputMap u = u ↔ u.a = zeroVec := by
  cases u
  simp [tiInputMap, vvToggle_eq_self_iff_zero]
```

2.2 The bridge point, stated as a single theorem

Combining the two facts gives the exact obstruction at the calibration bridge: on every nonzero VV column the reduced projection is preserved, yet the full input is not, because b moves.

```
/-- The exact calibration step cannot keep both the invariant projection and
the full post-switch input fixed on a nonzero VV column: `(z, a)` is preserved,
but `b` changes and hence the full input changes. -/
theorem calibrationFullInputPreservation_fails_of_nonzeroColumn
    {Z : Type*} {k : ℕ} (u : PostSwitchInput Z k)
    (ha : nonzeroColumn u.a) :
    invariantProjection (tiInputMap u) = invariantProjection u ∧
      tiInputMap u ≠ u ∧ (tiInputMap u).b ≠ u.b := by
  exact ⟨invariantProjection_tiInputMap u,
    tiInputMap_ne_self_of_nonzeroColumn u ha,
    tiInputMap_changes_b_of_nonzeroColumn u ha⟩
```

As a global no-go form: in every positive VV width and for every concrete $\mathbf{z}$-value, the proposition that T_i preserves the *full* local input on all nonzero columns is false.

```
theorem not_calibrationFullInputPreservation_on_nonzeroColumns
    {Z : Type*} (z : Z) {k : ℕ} (hk : 0 < k) :
    ¬ (∀ u : PostSwitchInput Z k, nonzeroColumn u.a -> tiInputMap u = u)
```

Why this voids the exchangeability at fixed u. The calibration of §1.2 derives the symmetry of the conditional law of (X_i, Y_i) *at a fixed* $\mathbf{u} = (\mathbf{z}, a_i, b)$ from the bijection T_i. But by (4) and `tiInputMap`, T_i sends the local input $\mathbf{u} = (\mathbf{z}, a_i, b)$ to $\mathbf{u}' = (\mathbf{z}, a_i, b \oplus a_i)$. When $a_i \neq 0$ these are *different* points of the local alphabet: the involution exchanges configurations across the two fibers $\{b\}$ and $\{b \oplus a_i\}$, not within the single fiber $\{b\}$ on which the calibration conditions. The symmetry it supplies is therefore a symmetry of the law conditioned on the reduced projection $(\mathbf{z}, a_i)$ — equivalently, averaged over the b-coordinate — and not of the law at fixed full $\mathbf{u}$. The exchangeability "$(X_i, Y_i) \mid \mathbf{u}$ is exchangeable" holds, via this involution, only on the locus $a_i = 0$.

The $a_i = 0$ fiber is vanishing. The column $a_i = Ae_i$ is the i-th column of the Valiant–Vazirani matrix A, whose columns are pairwise independent and marginally uniform in $\{0,1\}^k$ (v12, Definition 3.3). Hence $\Pr[a_i = 0] = 2^{-k}$, and with $k = c_1 \log m$ this is $m^{-\Theta(1)}$ — a polynomially vanishing fraction of columns. The promise-preserving exchangeability that the calibration step needs at fixed $\mathbf{u}$ is thus available only on a vanishing fraction of the VV columns that index the per-bit rules, exactly the regime the lower bound must control.

2.3 A finite success-domination counterexample

The same module records, in the smallest nontrivial setting, why "two instances that share a local input" cannot be calibrated by a $\mathbf{u}$-measurable rule: a global decoder that reads the full instance can strictly out-predict every rule that depends only on the shared local input. Take two equally weighted full instances (encoded by a Boolean) that map to the *same* nonzero one-bit post-switch input. A global decoder recovers the witness on both; any local decoder, forced to use only the common input, is a fixed guess and is correct on at most one.

```
theorem twoPointGlobalSuccessCount_eq_two :
    twoPointGlobalSuccessCount = 2 := by rfl

theorem twoPointLocalSuccessCount_le_one (guess : Bool) :
    twoPointLocalSuccessCount guess ≤ 1 := by
  cases guess <;> decide

/-- Finite domination counterexample: the global decoder succeeds on both
full instances, while every decoder depending only on the shared local input
succeeds on at most one. -/
theorem finite_successDomination_counterexample :
    twoPointGlobalSuccessCount = 2 ∧
      (∀ guess : Bool, twoPointLocalSuccessCount guess ≤ 1) ∧
      (∀ guess : Bool, twoPointLocalSuccessCount guess < twoPointGlobalSuccessCount)
```

This is a finite witness that success domination (3) cannot be obtained *from sharing* $\mathbf{u}$ *alone*: when two outcomes collapse to the same local input but differ in the truth, the comparator that is forced through $\mathbf{u}$ loses to the global decoder. It is the local, combinatorial shadow of the fiber-crossing exhibited above; the involution that would have justified calibration moves $\mathbf{u}$ off the fiber rather than within it.

3 Status and scope

We state the scope of this finding exactly.

What is established. The action of the promise-preserving involution T_i of (4) on the post-switch input $\mathbf{u}_i = (\mathbf{z}, a_i, b)$ of (1) is $b \mapsto b \oplus a_i$; it preserves the reduced projection $(\mathbf{z}, a_i)$ (theorem `invariantProjection_tiInputMap`) and preserves the full input $(\mathbf{z}, a_i, b)$ *if and only if* $a_i = 0$ (theorems `tiInputMap_eq_self_iff_zeroColumn` and `calibrationFullInputPreservation_fails_of_nonzeroColumn`); on every nonzero column the third coordinate b genuinely moves, so the full-input preservation invoked by the calibration argument fails (`not_calibrationFullInputPreservation_on_nonzeroColumns`). A finite two-point success-domination counterexample (`finite_successDomination_counterexample`) shows that a $\mathbf{u}$-measurable comparator cannot dominate a global decoder merely because they share the local input. All of this is machine-checked, without `sorry`, in `Mettapedia.Computability.PNP.PostSwitchInputObstruction`. The consequence is precise: the exchangeability of (X_i, Y_i) at a fixed $\mathbf{u}$ that the calibration step (v12, Lemma 4.8 / Appendix A.5, Lemma A.17) derives from T_i holds, via that involution, only on the $a_i = 0$ fiber, a vanishing $2^{-k} = m^{-\Theta(1)}$ fraction of VV columns. The local-comparator-dominates- global-decoder calibration is therefore not valid as stated.

What is not established. This is an obstruction to the calibration *as written*, not a refutation of its conclusion. We have machine-checked the *structural* fact that the cited involution does not fix the full local input off the zero-column fiber; we have *not* established that no $\mathbf{u}$-measurable comparator can dominate a global decoder on the ensemble $\mathcal{D}_m$. That stronger, measure-theoretic statement — a genuine anti-concentration / independence claim about the conditional law of (X_i, Y_i) given the full $\mathbf{u}$, integrated against the Valiant–Vazirani and masking laws — is exactly the difficulty the proof must confront, and it is left open here. In particular, a repaired argument might still recover a valid domination bound by conditioning only on the invariant projection $(\mathbf{z}, a_i)$ and supplying a separate mechanism for the b-coordinate, or by establishing the exchangeability at fixed $\mathbf{u}$ from some structure other than T_i. Nothing in this work rules out such a repair.

P **versus** NP **remains open.** We make no claim, in either direction, about P versus NP. The finding concerns the validity of one step in one proposed route, not the truth of the separation. Consistent with this, the formalization itself asserts no separation: the project's route-status record carries the verdict `claimsFinalSeparation` = `false`, and the module above proves a property of the calibration bridge, not a circuit or complexity-class lower bound.

4 What remains

The obstruction localizes the open question very precisely. The route's lower bound is sound *given* the calibration link (3); the calibration link is justified, in v12, by an exchangeability that the cited involution supplies only on a vanishing fraction of columns. Closing the route therefore requires one of the following, none of which is provided here.

- *Calibration on the reduced projection.* Re-derive the success-domination bound with the local rules conditioned on the T_i-invariant projection $(\mathbf{z}, a_i)$ rather than on the full $\mathbf{u} = (\mathbf{z}, a_i, b)$, and show that neutrality and sparsification still pin the resulting rules near $1/2$. This trades the calibration gap for a coarser local interface and must be checked to still feed the per-block bounds of v12, §5–§6.

- *Exchangeability at fixed* $\mathbf{u}$ *from new structure.* Exhibit a promise-preserving symmetry that genuinely fixes the full $(\mathbf{z}, a_i, b)$ while toggling X_i, or otherwise establish the anti-concentration of $(X_i, Y_i) \mid \mathbf{u}$ directly against the masking and Valiant–Vazirani laws — the measure-theoretic claim left open in §3.

- *A quantitative* b*-marginal control.* Bound the contribution of the b-coordinate to a global decoder's advantage, so that the fiber-crossing of T_i costs only $m^{-\Omega(1)}$ and can be absorbed into the slack of (3).

Each of these is a substantive analytic task on the masked-and-isolated ensemble. Until one is carried out, the Switching-by-Weakness route does not, as written, establish its distributional lower bound, because the calibration step on which that bound depends is not valid at fixed $\mathbf{u}$ on the bulk of the ensemble. The machine-checked content of this paper is exactly the boundary of that gap, and nothing beyond it.

Provenance

The manuscript discussed is Ben Goertzel, P $\neq$ NP*: A Non-Relativizing Proof via Quantale Weakness and Geometric Complexity*, version v12. The formalization is work of the MeTTapedia formalization effort; the Lean development is the module `Mettapedia.Computability.PNP.PostSwitchInputObstruction`, which carries no `sorry`. Notation in Sections 1–1.2 follows v12 (the ensemble $\mathcal{D}_m$, the masks H_m, the Valiant–Vazirani labels (a_i, b) with $a_i = Ae_i$, the SILS features $\mathbf{z}$, the post-switch input $\mathbf{u}_i = (\mathbf{z}, a_i, b)$, the involution T_i, and the polytime-capped weakness K^{poly}).